

6 Head and Neck Surgery

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6.0 Anatomy of the Neck

Surgically, it is useful to consider the neck anatomy in terms of compartments or levels and to consider fascial layers. The neck is covered by a layer of superficial fascia, and the deep cervical fascia can be divided into superficial, middle, and deep layers (**Fig. 6.1**). This is relevant in terms of spread of infection and lymphatic spread of cancer. The neck can be grossly divided into the anterior and posterior neck. Posteriorly, of course, are the cervical spine and paraspinal muscles. Anterolaterally, the neck contents are considered in terms of nodal levels I through VI (**Fig. 6.2**).

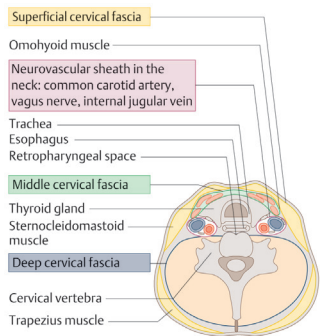


Fig. 6.1 Cervical fascial planes. (Used with permission from Probst R, Grevers G, Iro H. *Basic Otorhinolaryngology: A Step-by-Step Learning Guide*. Stuttgart/New York: Thieme;2006:313.)

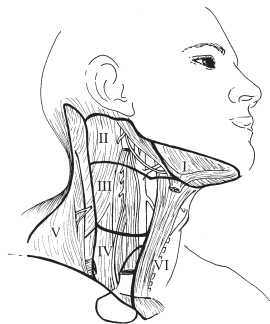


Fig. 6.2 Nodal levels I through VI. (Used with permission from Van de Water TR, Staecker H. *Otolaryngology: Basic Science and Clinical Review*. Stuttgart/New York: Thieme;2006:606.)

Each level contains a compartment of fibrofatty lymph node-bearing tissue that is addressed for removal during neck dissection procedures for cancer. Level I consists of the submental region and the submandibular area containing the submandibular gland. Level II is the upper jugular chain lymph node region, level III is the middle jugular chain, and level IV is the inferior jugular chain region. Level V is the posterior triangle from the anterior border of the trapezius to the posterior border of the sternocleidomastoid muscle. Level VI is the node-bearing tissue of the central compartment. The visceral structures of the neck are included in the central compartment and include the laryngotracheal complex, the thyroid and parathyroids, and the cervical esophagus.

◆ Blood Supply

The common carotid artery ascends into the neck and bifurcates into the internal and external carotid artery approximately at the level of the hyoid bone. The internal carotid has no branches in the neck and delivers important blood supply to the brain, along with the vertebral arteries. The external carotid supplies the head and neck structures via multiple branches (in order, starting inferiorly): the superior thyroid, ascending pharyngeal, lingual, facial, occipital, posterior auricular, superficial temporal, and maxillary arteries (**Fig. 6.3**). Venous drainage tends to parallel major arteries, with variable anterior and external jugular branches superficially, and large common facial veins joining into the internal jugular vein, which feeds the subclavian veins. The carotid sheath, formed by all three layers of the deep cervical fascia, contains the carotid artery, jugular vein, and vagus nerve. Important delicate lymphatics from the thoracic duct carry chyle into the jugular vein near its junction with the subclavian in the left neck; however, accessory thoracic ducts commonly occur on the right.

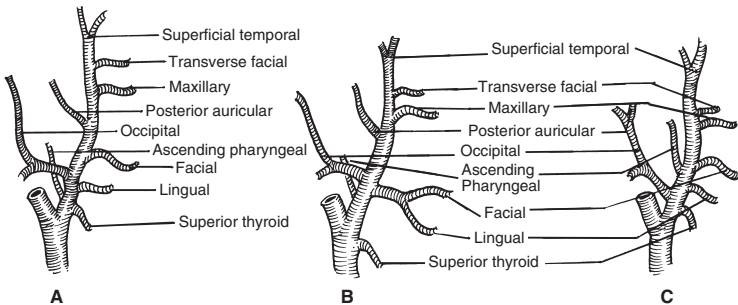


Fig. 6.3 The branching of the arteries originating from the carotid artery can be quite variable. (Used with permission from Van de Water TR, Staecker H. *Otolaryngology: Basic Science and Clinical Review*. Stuttgart/New York: Thieme;2006:603.)

◆ Innervation

The 12 cranial nerves (CNs) supply the head and neck region with motor, sensory, and special sensory fibers. They are, in order, the olfactory, optic, oculomotor, trochlear, trigeminal, abducens, facial, vestibulocochlear, glossopharyngeal, vagus, spinal accessory, and hypoglossal. In the neck there are surgically important aspects of CN anatomy. The hypoglossal (XII) emerges from the skull base at the hypoglossal canal and courses beneath the digastric and submandibular contents as motor supply to the tongue. The pars nervosa of the jugular foramen contains the glossopharyngeal (IX), vagus (X), and accessory nerves (XI). The vagus runs within the carotid sheath, and the recurrent laryngeal branches course in the tracheoesophageal groove to supply the vocal folds; the superior laryngeal nerves arise from the vagus in the upper neck. The accessory nerve courses posteroinferiorly through the neck, entering the anterior border of the sternocleidomastoid, emerging from the posterior border near Erb's point, and continuing into the trapezius.

The facial nerve (VII) exits the skull base at the stylomastoid foramen, and its main motor branches course within the parotid gland to innervate the muscles of facial expression, while the lingual nerve supplies parasympathetics to the submandibular gland and sensation to the anterior tongue and is encountered deep to the submandibular gland. Important cervical nerves are encountered in the neck as well. The great auricular nerve (it is not named greater, since there is no *lesser* auricular nerve) transmits sensory fibers of C2–3 to the auricular area (**Fig. 6.4**). This nerve is encountered along the sternocleidomastoid, coursing superiorly from Erb's point. The phrenic nerve (C3–5) is deep to the floor fascia (deep layer of the deep cervical fascia) along the anterior scalene muscle, supplying motor fibers to the diaphragm. Also, the brachial plexus is in the same plane, deep to the fascia on the middle scalene. The cervical sympathetic trunk runs deep to the carotid. CN anatomy is reviewed in **Appendix B** of this book.

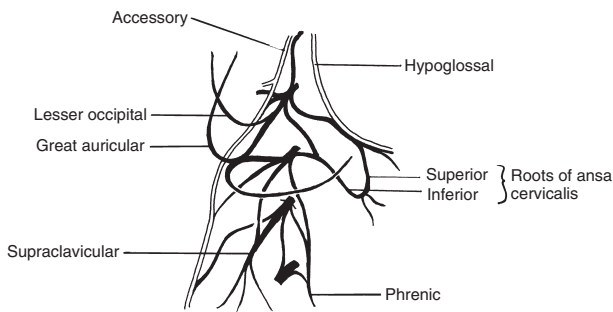


Fig. 6.4 Cervical plexus. (Used with permission from Van de Water TR, Staeker H. *Otolaryngology: Basic Science and Clinical Review*. Stuttgart/New York: Thieme;2006:603.)

6.1 Neck Emergencies

6.1.1 Necrotizing Soft Tissue Infections of the Head and Neck

◆ Key Features

- Necrotizing fasciitis is a soft tissue infection that causes necrosis of fascia and subcutaneous tissue but initially spares skin and muscle.
- It is rare in the head and neck; when it occurs it is usually associated with immunosuppression or odontogenic infection.
- Half of patients develop systemic bacteremia.
- Necrotizing soft tissue infections require aggressive treatment to combat the associated high morbidity and mortality.

Necrotizing soft tissue infections rarely may involve the face and neck, scalp, and eyelids.

◆ Epidemiology

In the head and neck, dental infections are the most common etiology, followed by trauma, peritonsillar and pharyngeal abscesses, and osteoradionecrosis. Immunocompromised patients are most susceptible. Predisposing conditions include diabetes mellitus, obesity, arteriosclerosis, alcoholism, chronic renal failure, hypothyroidism, malignancy, and poor nutrition. Gas-producing wound infections are usually produced by bacteria of the Clostridia class. Onset of symptoms is usually 2 to 4 days after the insult.

◆ Clinical

Signs

The signs of a necrotizing soft tissue infection include a low-grade fever, and the skin becomes smooth, warm, tense and shiny with no sharp demarcation. The infected skin also develops a dusky discoloration with poorly defined borders. Soft tissue crepitus is common from gas formation (though not necessary for the diagnosis). Later in the disease process, bullae may develop. Systemic symptoms get more severe with typical signs of septicemia. The last stage of the disease is characterized by cyanotic skin discoloration typical of necrosis. Patients may present with a benign clinical picture yet with a leukocytosis not consistent with the clinical picture.

Symptoms

There is sudden severe pain and swelling. There may be anesthesia of the involved skin.

Differential Diagnosis

The differential diagnosis includes other deep neck space infections, pyoderma gangrenosum, radiation necrosis, cellulitis, and erysipelas.

◆ Evaluation

Physical Exam

When clinical hard signs are present (i.e., crepitus, skin necrosis, bullae, hypotension), a physical exam may be helpful. Unfortunately, the signs and symptoms are often very subtle on presentation.

Imaging

Plain soft tissue films of the neck (looking for gas in soft tissue) may be obtained. Computed tomography (CT) is the most useful study to detect gas in areas inaccessible to palpation and to identify areas where the infection has spread. In addition, it can detect vascular thrombosis, erosion of vessels, and mediastinitis. Magnetic resonance imaging (MRI) has also been suggested, as it can demonstrate areas of enhancement or infection that crosses fascial planes.

Labs

Routine blood work is needed to look for metabolic abnormalities such as hyponatremia and hypoproteinemia due to fluid sequestration and hypocalcemia as a result of subcutaneous fat saponification. Additionally, a complete blood count will note a significantly elevated white blood cell count.

Microbiology

Group A hemolytic streptococci and *Staphylococcus aureus*, alone or in synergism, are frequently the initiating infecting bacteria. However, other aerobic and anaerobic pathogens may be present, including *Bacteroides*, *Clostridium*, *Peptostreptococcus*, Enterobacteriaceae, coliforms, *Proteus*, *Pseudomonas*, and *Klebsiella* species.

Pathology

The pathology shows a localized necrosis of skin that is secondary to thrombosis of nutrient vessels as they pass through involved fascia. Tissue hypoxia resulting from small vessel vasculitis and thrombi and impaired host defenses help facilitate anaerobic bacterial growth.

◆ Treatment Options

Prompt measures offer hope for survival of the patient.

Medical

High-dose intravenous (IV) broad-spectrum antibiotics are recommended. Hyperbaric oxygen therapy is sometimes used in addition to surgical and antimicrobial treatment.

Surgical

Aggressive surgical debridement until there are bleeding viable borders is necessary. This may require multiple trips to the operating room for irrigation and debridement and/or bedside dressing changes and debridement. Initially, it is important not to close the wound with larger defects, due to the progressive nature of the disease process. If the necrotizing process involves the neck, avoid tracheotomy through the infected area.

◆ Outcome and Follow-Up

The overall mortality rate has been reported to be as high as 70%.

6.1.2 Ludwig's Angina

◆ Key Features

- Ludwig's angina is a rapidly expanding, diffuse inflammation of the submandibular and sublingual spaces.
- It is most often caused by dental infections.
- The condition is often found in immunocompromised patients, such as those with diabetes or human immunodeficiency virus/acquired immunodeficiency syndrome (HIV/AIDS) or drug abusers.

Ludwig's angina is a rapidly spreading bilateral cellulitis of the sublingual and submaxillary spaces. Before the advent of antibiotics, the mortality associated with Ludwig's angina approached 50%, but today, mortality rates are in the range of 8 to 10%. The most common cause of death is respiratory compromise.

◆ Epidemiology

Ludwig's angina represents up to 13% of all deep neck infections. Typically, young adults present with Ludwig's angina; it is unusual in children. The infection generally spreads from a dental or periodontal infection. Other causes include upper respiratory infections, floor-of-mouth trauma, mandibular fractures, sialadenitis, IV drug abuse, trauma and tonsillitis, and immunocompromised states such as diabetes or HIV/AIDS. Forty percent of Ludwig's angina cases involve oral anaerobes.

◆ Clinical

Signs

The signs of the infection include an inability to close the mouth, trismus, drooling, sitting upright, inability to swallow and dysphonia, dyspnea,

stridor, fever, chills, and tachycardia. Note that stridor, dyspnea, decreased air movement, or cyanosis suggest impending respiratory compromise.

Symptoms

The symptoms of Ludwig's angina are severe neck pain and tenderness, submandibular and submental swelling, fever, malaise, and dysphagia.

Differential Diagnosis

The differential diagnosis includes other deep neck space infections, an infected cyst, tumor, and cellulitis.

◆ Evaluation

Physical Exam

On physical examination, the patient will often present with carious molar teeth, neck rigidity, and drooling. There is a "woody" or "brawny" induration of involved spaces with little to no fluctuance. The patient's mouth floor is swollen, and the tongue may be swollen or elevated. The patient may describe difficulty breathing and may have audible breathing.

Imaging

A computed tomography (CT) scan is most useful.

Labs

Blood cultures are usually negative; if the swelling is aspirated or drained, samples should be sent for Gram stain, culture, and sensitivity.

◆ Treatment Options

Airway control is the first priority of treatment, followed by IV antibiotics and timely surgical drainage. Blind oral or nasotracheal intubation or attempts with neuromuscular paralysis are contraindicated in Ludwig's angina, as they may precipitate an airway crisis.

Medical

Aggressive empiric high-dose, IV antibiotics are recommended—cefuroxime plus metronidazole. If the patient is allergic to penicillin, prescribe clindamycin plus a quinolone. Once culture and sensitivity results have been obtained, antibiotic therapy may be changed accordingly.

Surgical

To establish airway control, a tracheotomy may be indicated. Surgical drainage was once universally required but now may be reserved for cases in which antibiotic treatment fails. On external incision and drainage, often straw-colored material, as opposed to frank purulence, is found. Consider placement of a passive drain to allow continued drainage as the infection resolves.

◆ Complications

A spontaneous rupture may lead to asphyxia, aspiration, or pneumonia. The infection may spread to other deep neck compartments.

◆ Outcome and Follow-Up

The offending tooth should also be removed if the infection's origin is odontogenic.

6.1.3 Deep Neck Infections

◆ Key Features

- Deep neck infections are most commonly caused by tonsillar, peritonsillar, or odontogenic infections.
- They may involve surrounding nerves, vessels, bones, and other soft tissue.
- Microbiology typically reveals mixed bacterial flora, including anaerobic species.
- Deep neck spaces have avenues of communication with each other: infection in one space can spread to adjacent spaces.

Deep neck infections hold the potential for severe complications. Complex head and neck anatomy often makes early recognition of deep neck infections challenging, and a high index of suspicion is necessary to avoid delay in treatment. Aggressive monitoring and management of the airway are the most urgent aspects of care, followed by appropriate antibiotic coverage and surgical drainage, as needed. Risk factors for deep neck infection include diabetes mellitus, HIV, steroid therapy, chemotherapy, and other sources of immune compromise.

◆ Clinical

Signs and Symptoms

Pain and swelling of the neck are the most prevalent symptoms. Fever, malaise, and dysphagia may occur. Other common symptoms are deep space-specific and include dysphagia, odynophagia, trismus, dysphonia, otalgia, and dyspnea.

In the pediatric population, fever, neck mass, and stiffness are most prevalent, followed by sore throat, poor oral intake, drooling, and lymphadenopathy. Stridor, dyspnea, decreased air movement, and cyanosis suggest impending respiratory compromise.

Differential Diagnosis

The differential diagnosis includes infected congenital cysts, lymphangitis, tumor, cellulitis, and necrotizing fasciitis.

◆ Evaluation

History, physical examination, laboratory work, and diagnostic imaging each provide important clues for assessing a patient for a deep neck infection.

Physical Exam

Initial evaluation of the airway is always the first priority, and any signs of respiratory distress or impending airway compromise should be immediately and aggressively managed.

Imaging

A contrast-enhanced computed tomography (CT) scan is most useful. Ultrasound may be more accurate than CT in differentiating a drainable abscess from cellulitis. Magnetic resonance imaging (MRI) provides better soft tissue definition than CT. MRI also avoids exposure to radiation and interference from dental fillings.

Labs

Labs should include a complete blood count (CBC) with differential, serum glucose, and electrolytes; coagulation studies; HIV screening in adults; blood cultures; and appropriate cultures of aspirates obtained before antibiotics are instituted, if possible.

Microbiology

Infections are commonly polymicrobial and reflect the oropharyngeal flora. Frequently isolated aerobes include *Streptococcus viridans*, *Klebsiella pneumoniae*, *Staphylococcus aureus*, and, less frequently, *Streptococcus pneumoniae*, *Streptococcus pyogenes*, *Neisseria* species, and *Haemophilus influenzae*. Common anaerobic isolates include *Peptostreptococcus*, *Bacteroides fragilis*, pigmented *Prevotella* and *Porphyromonas* spp., *Fusobacterium* spp., and *Eikenella corrodens*.

◆ Treatment Options

Airway control is the first priority of treatment, followed by IV antibiotics and timely surgical drainage. Airway management, if necessary, should be undertaken under controlled conditions, in the operating room if possible, with either awake fiberoptic intubation or awake tracheotomy. Blind oral or nasotracheal intubation or attempts with neuromuscular paralysis may precipitate an airway crisis.

Medical

Every patient who has a deep neck infection should be given empiric antibiotic therapy until culture and sensitivity results are available. Empiric

therapy should be effective against the aerobic and anaerobic bacteria that are commonly involved.

Either penicillin in combination with a β -lactamase inhibitor (e.g., amoxicillin with clavulanic acid) or a β -lactamase-resistant antibiotic (e.g., cefoxitin, cefuroxime, imipenem, or meropenem) in combination with a drug that is highly effective against most anaerobes (e.g., clindamycin or metronidazole) is recommended for optimal empiric coverage. Once available, the results of the culture and sensitivity tests can enable tailoring of adequate antibiotic therapy.

In select cases, an uncomplicated deep neck abscess or cellulitis can be effectively treated with antibiotics and careful monitoring, without surgical drainage. Simultaneous medical treatment for associated comorbidities such as diabetes mellitus can improve the overall immune status of a patient.

Surgical

Indications for surgery include airway compromise, critical condition, septicemia, complications, descending infection, diabetes mellitus, or no clinical improvement within 48 hours of the initiation of parenteral antibiotics. In addition, abscesses > 3 cm in diameter that involve the prevertebral, anterior visceral, or carotid spaces, or that involve more than two neck spaces, should be surgically drained.

Surgical drainage (**Table 6.1**) can be performed in several ways, including simple intraoral or extraoral incision and drainage for superficial abscesses, a more extensive external cervical approach with drain placement for more complicated infections, and minimally invasive techniques such as image-guided needle aspiration and indwelling catheter placement.

Table 6.1 Surgical approaches for drainage of deep neck infections

Infection site	Surgical approaches for drainage
Peritonsillar	Intraoral needle aspiration or incision and drainage
Submandibular space	Supramylohyoid— <i>intraoral</i> drainage Inframyllohyoid— <i>extraoral</i> surgical drainage
Parapharyngeal space	External cervical approach along the anterior border of the sternocleidomastoid muscle CT-guided transoral drainage
Masticator space	External incision along inferior border of mandible Intraoral approach via retromolar trigone
Parotid space	External parotidectomy incision
Retropharyngeal space	Transoral External cervical approach
Prevertebral space	External cervical approach
Carotid space	External cervical approach

◆ Complications

Complications include mediastinitis, aspiration pneumonia, lung abscess, empyema, Lemierre's syndrome (suppurative thrombophlebitis of the internal jugular vein), carotid artery aneurysm or rupture, osteomyelitis involving the mandible or cervical vertebral bodies, meningitis, intracranial abscess, and disseminated intravascular coagulation.

◆ Outcome and Follow-Up

The initiating etiology, if recognized (an infected tooth, a tonsillar abscess), and predisposing systemic conditions (diabetes mellitus) should be addressed.

6.1.4 Neck Trauma

◆ Key Features

- Neck trauma can be either penetrating or blunt.
- There is potential injury to the larynx, trachea, esophagus, major vessels, and nerves.
- Airway management is always the priority.
- Cervical spine injury must be excluded.
- The zone of the neck where the trauma occurs will help dictate the management algorithm.

Factors in the mechanism of neck trauma determine the location of injury, the injury characteristics, the tissues and organs involved, and the extent of damage to the tissues and organs.

◆ Epidemiology

Neck trauma accounts for 5 to 10% of all serious traumatic injuries. Blunt trauma to the neck typically results from motor vehicle crashes, but it also occurs with sports-related injuries, clothesline injury, strangulation, or blows from the fists or feet. Blunt trauma has become much less common since routine seatbelt use was established.

For penetrating trauma, > 95% of wounds result from guns and knives, with the remainder resulting from motor vehicle accidents, household injuries, industrial accidents, and sporting events. The male-to-female ratio of penetrating neck trauma is 5:1.

◆ Clinical

Critical organs and structures are at risk from neck trauma; clinical manifestations may vary greatly. The presence or absence of signs and symptoms

can sometimes be misleading, serving as a poor predictor of underlying damage.

Signs

Signs of airway injury include:

- Subcutaneous emphysema—tracheal, esophageal, or pulmonary injury
- Air bubbling through the wound
- Stridor or respiratory distress—laryngeal and/or esophageal injury
- Voice changes: hoarseness, diplophonia, breathiness
- Cyanosis

Signs of vascular injury include:

- Hematoma (expanding)—vascular injury
- Active external hemorrhage from the wound site—arterial vascular injury
- Bruit/thrill—arteriovenous fistula
- Pulselessness/pulse deficit
- Distal ischemia (neurologic deficit in this case)

Signs of pharyngoesophageal injury include:

- Hematemesis, inability to tolerate secretions
- Neck crepitus
- Development of mediastinitis

Symptoms

- Clinical manifestations may vary greatly depending on involved organs and systems.
- Dysphagia—tracheal and/or esophageal injury
- Hoarseness—tracheal and/or esophageal injury
- Oronasopharyngeal bleeding—vascular, tracheal, or esophageal injury
- Neurologic deficit—vascular and/or spinal cord injury
- Hypotension—nonspecific; may be related to the neck injury or may indicate trauma elsewhere

Differential Diagnosis

Considerations with neck trauma include cervical spine injury, laryngotracheal injury, vascular injury, and pharyngoesophageal injury.

◆ Evaluation

History

History, if available, can provide important details regarding the mechanism of injury.

Physical Exam

Advanced Trauma Life Support (ATLS) protocols are followed. The exam begins with ABCs (airway, breathing, circulation), followed by secondary assessment once a patient has a safe airway and is hemodynamically stable. All patients with neck trauma should be assumed to have a cervical spine injury until this has been ruled out.

With blunt trauma, injury to the larynx or trachea is the most common serious finding and often presents with subcutaneous air, hoarseness, or odynophagia. In a stable patient, flexible fiberoptic laryngoscopy can reveal evidence of injury such as blood, motion impairment, or edema.

With penetrating trauma, determine which vertical zones of the neck are involved (**Fig. 6.5**; **Table 6.2**). Zone I extends from the clavicle to the cricoid cartilage; zone II extends from the cricoid to the mandibular angle; zone III extends from the mandibular angle to the skull base.

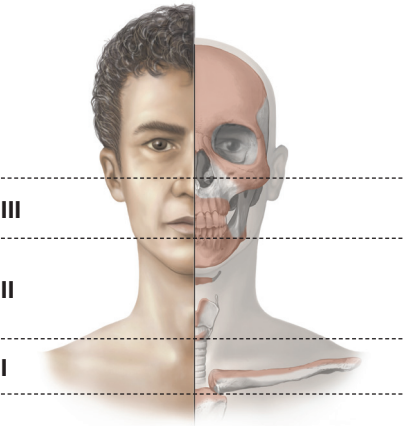


Fig. 6.5 Zones of the neck for evaluation of penetrating trauma. (Used with permission from Schuenke M, Schulte E, Schumacher U. *Head, Neck, and Neuroanatomy*. 2nd ed. Stuttgart/ New York: Thieme; 2016. *Thieme Atlas of Anatomy*; vol 3, illustration by Karl Wesker.)

Table 6.2 Zones of the neck and management of penetrating trauma

Zone	Anatomy	Contents	Management
I	Clavicle to cricoid	Common carotid, vertebral, and subclavian arteries and the trachea, esophagus, thoracic duct, and thymus	Angiography, esophagram
II	Cricoid to mandibular angle	Internal and external carotid arteries, jugular veins, pharynx, larynx, esophagus, recurrent laryngeal nerve, spinal cord, trachea, thyroid, and parathyroids	Surgical exploration if symptomatic
III	Mandibular angle to skull base	Distal extracranial carotid and vertebral arteries and the uppermost segments of the jugular veins	Angiography

Imaging

Controversy exists regarding aspects of trauma management, with trends moving away from surgical exploration in stable patients and with expanding use of imaging and observation. There is an inherent delay with any imaging study; accordingly, transport to the operating room should not be delayed by an imaging study when the patient's condition warrants emergent surgery.

Penetrating zone I and III injuries should undergo angiographic evaluation of the carotids. Zone I injuries should also undergo esophageal studies such as Gastrografin (Bracco Diagnostics, Inc., Princeton, NJ) swallow study.

Cervical spine radiography is routine and should be reviewed for emphysema, fractures, displacement of the trachea, and the presence of a foreign body.

Computed tomography (CT) scans prove most useful when bony or soft tissue damage is a consideration. Magnetic resonance imaging (MRI) or MR angiography is used for evaluation of the patient exhibiting neurologic impairment with minimal or absent abnormalities of the cervical spine on plain radiograph.

CT angiography offers advantages over conventional catheter angiography. It is readily accessible, can be rapidly performed, and causes fewer complications than catheter angiography. Additionally, some experts assert that subtle disruptions of the vessel wall may be detected on CT angiography. Artifacts secondary to metal can obscure vascular detail and can limit CT angiography.

Labs

As necessary, a trauma panel, CBC, electrolytes, other warranted blood chemistry levels, and blood type and crossmatching should be obtained.

◆ Treatment Options

Establishment of the airway is the first priority. If there is airway compromise, a surgical airway, rather than endotracheal intubation, is usually preferred. With neck injuries, a patient may have a laryngotracheal separation or near separation that can be worsened by an endotracheal intubation. Either a cricothyroidotomy or tracheotomy is performed if there is respiratory distress. Bleeding is initially managed with direct pressure and establishment of large-bore IV access to permit fluid resuscitation (**Fig. 6.6**).

- Endoscopy
 - Laryngoscopy, bronchoscopy, pharyngoscopy, and esophagoscopy may be useful in the assessment of the aerodigestive tract. Rigid endoscopes are superior to flexible scopes, as insufflation may worsen an injury.
- Angiography
 - Angiography is routinely used to evaluate stable patients sustaining penetrating wounds to zones I and III that pierce the platysma.
 - A four-vessel study is a prerequisite.
 - Drawbacks include cost and the inherent danger of any vascular, particularly arterial, invasive procedure.

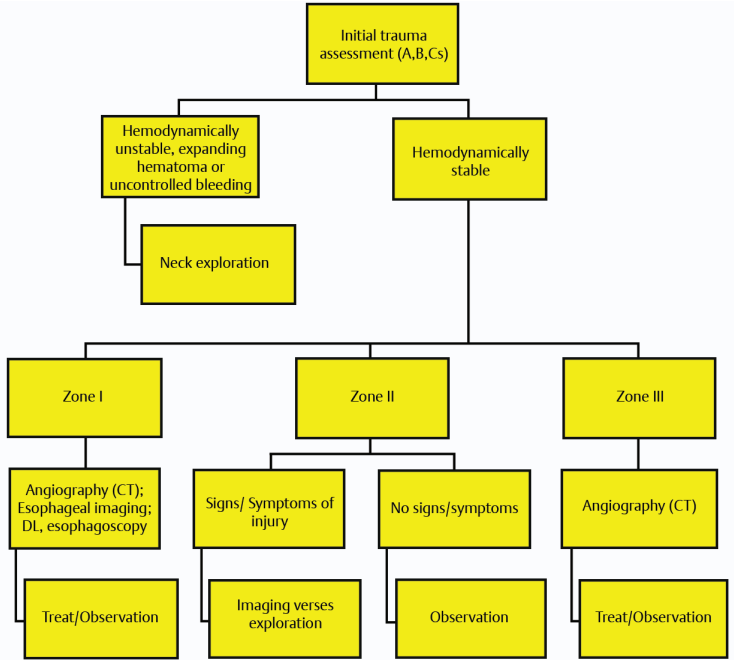


Fig. 6.6 Neck trauma management algorithm.

The unstable patient (hemodynamic instability, severe hemorrhage, expanding hematoma) is taken to the operating room. The stable patient is categorized as symptomatic or asymptomatic. Signs or symptoms of injury to the airway, esophagus, vessels, or nerves (i.e., hemoptysis, hoarseness, dysphagia, crepitus) dictate further work-up/interventions. If the penetrating injury is in zone II, neck exploration is generally done; if the injury is zone I, angiography and esophageal studies are performed. If the injury is in zone III, angiography should be performed. Asymptomatic patients are also typically imaged as above if the injury is zone I or III and observed if the injury is zone II.

In general, vascular injuries are managed with either embolization or surgical control. Surgery involves exploration and management of injuries of the carotid sheath, esophagus, and laryngotracheal complex (**Table 6.3**).

Table 6.3 Specific injuries sought and treated during neck exploration

Carotid artery injuries	Esophageal injuries
Vertebral artery injuries	Nerve injuries
Jugular vein injury	Thoracic duct injuries
Laryngotracheal injuries	Thyroid injuries

There is no role for probing or local exploration of the neck in the trauma bay or emergency room, because this may dislodge a clot and initiate uncontrollable hemorrhage.

◆ Outcome and Follow-Up

Standard postoperative management for neck surgery is followed. Overall mortality is 1 to 2%.

6.2 Approach to Neck Masses

◆ Key Features

- A neck mass may be inflammatory, congenital, or neoplastic.
- It may be anterior (midline), lateral, posterolateral, or supraclavicular.
- The patient's history should include age, duration, progression, pain, infection, smoking, prior cancer, exposure to tuberculosis, and animal exposure.

Neck masses are relatively common and may present at any age group. The differential diagnosis is broad, and both benign and malignant processes should be considered. A systematic approach is crucial to developing a rapid diagnosis and treatment plan.

Age is an important factor in evaluating a neck mass. The age groups include pediatric, young adult, and older adult (> 40 years). Each age group exhibits a certain relative frequency of disease occurrences, which can guide the diagnostician to further differential considerations. In general, neck masses in children are more commonly inflammatory or infectious. This is also true in young adults, although lymphoma is a consideration. In older adults, a neck mass should be considered neoplastic until proven otherwise.

The location of congenital neck masses is important; these lesions are often characterized by their location. The location of malignant neck masses, particularly if metastatic, may help identify the primary tumor (**Fig. 6.7**).

◆ Clinical

Signs and Symptoms

Depending on the cause, the neck mass may be painless (early neoplasm or congenital mass) or painful (infection or trauma). Depending on the etiology, associated symptoms may be those of an upper respiratory infection, toothache (infectious or inflammatory mass), or dysphagia, odynophagia, hoarseness, otalgia, hemoptysis, weight loss, night sweats, and fever (neoplasm).

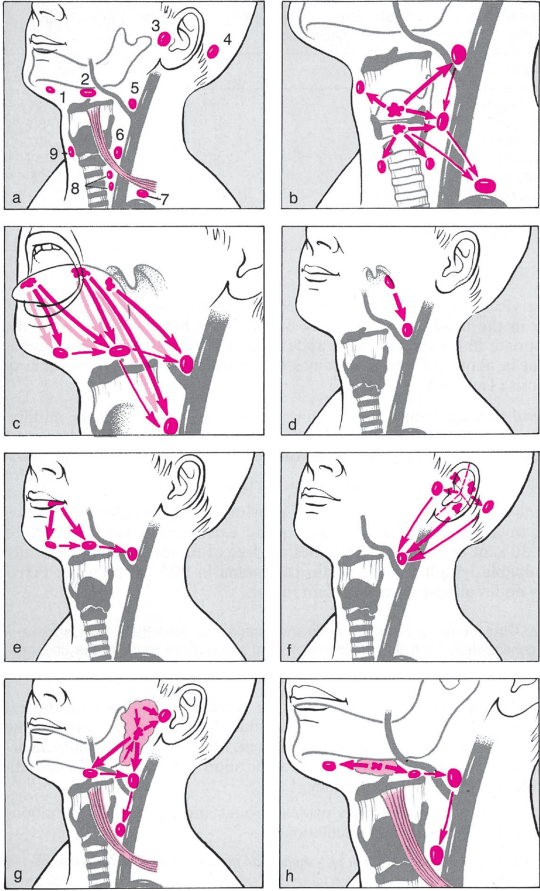


Fig. 6.7 Head and neck masses. **(a)** Typical sites of regional lymph node metastases: 1. submental lymph nodes; 2. submandibular lymph nodes; 3. parotid and preauricular lymph nodes; 4. retroauricular lymph nodes; 5. lymph nodes of the jugulofacial venous angle; 6. deep cervical lymph nodes; 7. lymph nodes in the juguloclavicular venous angle: lower deep cervical lymph nodes and supraclavicular lymph nodes; 8. pre- and peritracheal lymph nodes; 9. prelaryngeal lymph nodes. **(b)** Laryngeal carcinoma. **(c)** Carcinoma of different parts of the tongue. Note the tendency to contralateral metastases. **(d)** Tonsillar carcinoma. **(e)** Lower lip carcinoma. **(f)** Carcinoma of the external ear. Note the segmental lymphatic efferent from the auricle. **(g)** Parotid carcinoma. Note the intraglandular lymph node metastases. **(h)** Submandibular gland carcinoma. (Used with permission from Becker W, Naumann HH, Pfaltz CR. *Ear, Nose and Throat Diseases: A Pocket Reference*. 2nd ed. Stuttgart/New York: Thieme;1994:516.)

Differential Diagnosis

See **Table 6.4**.

Table 6.4 Differential diagnosis of neck masses

Congenital	Neoplastic malignant	Neoplastic benign	Systemic, inflammatory and infectious disease
Branchial cleft cyst	Head and neck metastasis	Hemangioma	Lymphadenitis—bacterial, viral
Dermoid cyst	Thyroid metastasis	Thyroid: neoplasm or goiter	Deep neck abscess
Thyroglossal duct cyst	Lymphoma	Lymphangioma	Cat scratch disease
Laryngocele	Salivary gland tumor	Salivary gland tumor	Tuberculosis or atypical mycobacterium
Ectopic thyroid tissue		Lipoma	Sarcoidosis, Kikuchi-Fujimoto’s disease, Kimura’s disease, Castleman’s disease
		Carotid body tumor or paraganglioma	Sialadenitis

◆ **Evaluation**

History

A thorough review of the developmental time course of the mass, associated symptoms, personal habits prior to the trauma or infection, irradiation, or surgery is important. Ask about smoking, tobacco chewing, alcohol use, fever, pain, weight loss, night sweats, exposure to tuberculosis, animals, pets, and occupational/sexual history.

Physical Exam

All mucosal surfaces of the nasopharynx, oropharynx, larynx, and nasal cavity should be visualized by direct examination or by indirect mirror or fiberoptic visualization. State of dentition should be noted. The oropharyngeal surfaces should be palpated. Regarding the neck mass, emphasis on location, tenderness, mobility, and consistency of the neck mass can often place the mass within a general etiologic grouping.

Imaging

Computed tomography (CT) is important and can better elucidate the nature of a neck mass and its association with surrounding structures. Criteria such as heterogeneity of the center of the mass, blurred borders, and a

round shape are suggestive of malignancy. CT may also be able to identify additional neck masses not clinically palpable and/or the primary tumor. Contrast should be used *except* in a suspected thyroid lesion, as it may interfere with radioactive iodine imaging studies or therapy.

Magnetic resonance imaging (MRI) is comparable to CT, but it is more expensive and has longer imaging time. With contrast, it is especially good for vascular and neural delineation. Angiography is recommended if a primary vascular neck mass is suspected and synchronous embolization is being considered. Ultrasound is helpful in differentiating solid from cystic masses and congenital cysts from solid lymph nodes and glandular tumors.

Labs

A routine complete blood count test may reveal an infectious process. More specialized laboratory tests may become necessary as the investigation proceeds.

Other Tests

- Fine-needle aspiration biopsy (FNAB): Currently, FNAB is the standard of diagnosis for neck masses and is indicated in any neck mass that is not an abscess and persists despite appropriate antibiotic therapy. FNAB separates inflammatory and reactive processes that usually do not require surgery from neoplastic lesions, either benign or malignant.
- Panendoscopy: If careful examination in the office does not identify the etiology of the neck mass and a tumor is suspected, the upper aerodigestive tract should be examined under anesthesia. Biopsies should be performed on any suspicious mucosal lesions.

◆ Treatment Options

Medical

A tender, mobile mass or one highly suggestive of inflammatory or infectious etiology may warrant a short clinical trial of antibiotics and observation with close follow-up. Use steroids judiciously; steroids may shrink a neck mass caused by lymphoma, lulling the physician and patient into a false sense that the condition is improving.

Surgical

- Open excisional biopsies should be avoided in cases in which a nonlymphoma malignancy (epidermoid, melanoma) is suspected. If FNAB results are negative or equivocal but suspicion for malignancy persists, an open excisional biopsy of the cervical lymph node may be performed. The patient and surgeon should be prepared to proceed immediately with a complete neck dissection depending on the results of frozen sections.
- Open excisional biopsies may be performed for lymphomas or granulomatous disease.
- Inflamed congenital masses are typically treated with antibiotics and then surgically removed after inflammation has subsided.

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